E350

U1300 and U3400

RELEASE NOTES FOR F/W VERSION R5.4

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Revision history

Version	Date	Comments
1	30/11/2018	Initial Release

Important Information
This document provides information relating to E350 U1300 Series 1, Series 2 and U3400 meter firmware (the "Product")

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1 About this document

1.1 Purpose & Scope

The purpose of this ***R5.4 Firmware Release Note*** is to communicate new features, limitations and changes in meter functionality.

The R5.4 firmware release is for the E350 - U1300 and U3400 meters.

1.2 Definitions, Acronyms, and Abbreviations

RSM:	Relay State Machine for the load control relay
LC:	Load Control
LP:	Load Profile
CLC/R:	Controlled Load Contactor or Relay
CSM:	Customer Supply Monitoring including Neutral Integrity
QoS:	Quality of Supply
AMI:	Advanced Metering Infrastructure
AD:	Auto Disconnect
S2:	Series 2
SCC:	Supply Capacity Control
STS:	Smart Time Synchronisation
HES:	Head End Systems
E350:	Family name of Landis+Gyr modular meter including the types U1200, U1300, U1300 S2 U3300, U3400 and U3350
E650:	Family name of Landis+Gyr commercial and industrial meter including the type EM5400 and EM5400 S2
E420:	Family name of Landis+Gyr integrated meter S1100 RF and S1100 LTE
R1100:	Landis+Gyr’s smart load control device.

1.3 References

E350 U1300 User Manual

E350 U3400 User Manual

EMPWin+ User Manual

2 About the release, compatible products and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released firmware version: R5.0

This Released Version: R5.4

Compatible Meter	Firmware Upgrade*	Firmware Downgrade*
U1300 Series 1	4.2 → 5.4 ↓ 4.3 → 5.4 ↓ 4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.8 → 5.4 ↓ 4.9 → 5.4 ↓ 5.0 → 5.4	5.4 → 5.0
U1300 Series 2	4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.8 → 5.4 ↓ 4.9 → 5.4 ↓ 5.0 → 5.4	5.4 → 5.0
U3400	4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.7 → 5.4 ↓ 4.9 → 5.4 ↓ 5.0 → 5.4	5.4 → 5.0

Important Note: *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please also refer to the intermediary firmware release notes when skipping meter firmware version.

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1 and Series 2)
- E350 U3400 meters
- EMPWin+ 3.4.4

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The FT100 release 5.1.2 or earlier must not be used with R5.0 or R5.4 meter firmware.

EMPWin+ 3.3.3 or greater must be used with R5.0 & R5.4 meter firmware.

E350 U1300 S2 meters cannot be downgraded to any previous R4.3 or earlier F/W Version.

2.1 New Features for Supported Meters

2.1.1 General New Features

- The E350 U1300 & U3400 meters with R5.4 firmware and has the following enhancements.
  - The U1300 R5.4 functionality is the same as the r5.0 firmware but with capital ‘R’ in the firmware version numbering to be compatible with SSN’s UIQ system.
  - The U3400 R5.4 functionality is the same as the R5.0 firmware but with the tamper feature enabled. The U3400 tamper feature was inadvertently disabled in the U3400 R5.0 firmware.

2.1.2 Firmware Upgrade

- The E350 U1300 & U3400 meters can be F/W upgraded from any R4.x to R5.0.
- The LP data is maintained when the firmware upgrade process expands the available LP memory. However, a reprogram is required to utilise this memory.
- The TOU data is maintained when the firmware upgrade process expands the TOU area from 8 to 35 days of self-reads. Previous Billing data can be corrupted when F/W is upgraded from any R4.x to R5.0 or R5.4. In most applications the Previous Billing data is not used by HES systems or by customers. Re-programming the meter is required to enable the new functionality provided in R5.0 or R5.4 and will also clear the Previous Billing data.

2.1.3 Firmware Downgrade

Downgrading from R5.4 to R5.0 is supported. However, downgrade from R5.4 to R4.x is not supported due to a hardware flag being automatically set when the meter is upgraded to R5.0. or R5.4 This cannot be changed by the customer when downgrading the F/W back to R4.x. If a downgrade is attempted, Customers will notice a program read can’t be performed (with EMPWin+ version 3.3.3 or greater) and if an earlier EMPWin+ version were used, they will see the TOU and LP data will become corrupted. This is due to the structural changes required in order to support 35 TOU self-read periods and additional LP area that was created during the R5.0 or R5.4 F/W upgrade.

2.1.4 Billing Self Read Periods

- E350 U1300 & U3400 meters that are upgraded to R5.0 or R5.4 F/W will automatically start recording 35 self-read periods instead of 8 as occurs in earlier F/W versions.

2.1.5 Load Profile

- The E350 U1300 & U3400 with R5.0 or R5.4 firmware now supports 5 minute Load profile interval selection and increased memory referred to as “Extended 1”. The extended memory can store up to 220 days of 5 minute LP intervals for a single LP group with four channels enabled.
- Meter configuration (MCD) files with 5 minute intervals enabled are restricted in MeterComm and MCDExport from being written to meters with older firmware.

2.2 Itron MV90, MVRS, IEE Support

MV90 has been tested with R5.4 F/W

- MV90 reads the whole 35 self-read records but will only report the most recent 8.
- MV90 can support the reading of 5 minute LP data.
- HHF files created by MeterViewer can be imported into MV90 with 5 minute LP data.

2.3 Command Center

2.3.1 Command Center Software Quality Assurance testing.

Command Center (CC) version 7.3.0.6 has been tested with R5.4 F/W.

- The meter F/W can be sent Over the Air (OTA) using CC.
- R4.9 and R4.8 to R5.4 was tested using RF comms 14.17.001
- U1300 OTA R4.8 to R5.4 Cellular comms 14.17.001 was OK.
- U3400 meter F/W upgrade R4.8 to R5.4 was tested using RF comms 14.11.001
- U3400 R5.4 comms F/W RF upgraded from 14.11.001 to 14.19.001.
- U3400 meter F/W upgrade OTA R5.0 to R5.4 Cellular comms 14.19.001.
- Meter continued to push LP, Billing Reads and Events to CC.
- CC can perform a complete read of the 35 self-read periods if read from Standard Table 26 and Interpret Known Registers is selected. It decodes and shows the oldest record only.
- CC can read LP data from meters that have 5, 10, 15, 30 or 60 minute LP intervals. Each LP group needs its own LP Set read command in its defined Configuration Group.
- The new Smart Time Synchronisation (**STS**) method in R5.0 & R5.4 F/W is compatible with CC methods of adjusting time.

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2.3.2 Command Center Solution Integration Testing.

Command Center (CC) version 7.2.0.8 has been tested with R5.4 F/W.

The following functionality has been verified through CC UI.

- Auto Registration
- Meter F/W upgrade through OTA
  - VAG Immediate
  - Broadcast Immediate (W/O UDAG)
- Module F/W upgrade through OTA
  - P2P Delayed
- Meter Modify Program:
  - Partial Programming (VAG Immediate) Through GSIS
  - Full Programming (P2P Delayed)
- Time Sync
- Midnight Energy Snapshots
- Load Profile Group Settings
- Power Outage

Solution Integration Test Environment

S.#	Servers	HDD (GB)	RAM (GB)	vCPU	OS	Build Version Series
1	CC Application Server (Web Server)	80	8	2 X 2 (4)	Windows Server 2012 R2 (64-Bit)	7.2.0.8
2	NMS	10	2	2 X 1	Ubuntu 14.04 LTS/SLES 6.6	1.3.3-0
3	DB (SQL)	80	8	2 X 2 (4)	Server 2012 R2 (64-Bit)/Oracle 12c	SQL Server 2008
4	SBS	120	16	2 X 1	Windows Server 2012 R2 (64-Bit)	1.0.23.6
5	PANA	80	4	2 X 1	Windows Server 2012 R2 (64-Bit)	1.0.0.9
6	GSIS	100	4	2x1	Windows Server 2012 R2 (64-Bit)	7.2.0.1

Device Type	Model	Firmware Version
Meters	U1300, U3400	05.05.R5.4
Communication Module	E35CRF	11.11
Collector App	BOTM	2.6.14
Collector Radio		10.79

2.4 Bug fixes in this release

Item No.	*Description*
TFS-11024 **TFS-11049**	A bug has been found on R5.0 that tamper functionality has been disabled for U3400 meter. This has been fixed in R5.4 F/W.
TFS-1066 **TFS-11048**	It was found in some U3400 meters, the self-check error functionality did not work correctly due to the Ripple Log wrapping combined with a power outage. Improvements have been made to Self Check error functionality. The power fail backup of the Ripple Event log has been improved so a self-check error should not occur.

2.5 Known Issues in this release

Item No.	*Description*
Mantis-3163	If there is a long power outage across a Season change date, then the Billing report might incorrectly label the season number as 0. The date logged, and energy/demand values are correct. This is an existing bug from previous firmware versions.
Mantis-3584	If there is a power outage across a LC season change from LC season 1 to LC season 0 then the event that indicates this LC season change to 0 is not recorded. The logging works correctly for LC season change from 0 to 1 across a power outage. The LC scheduler works correctly it is only the event that is missing.
Mantis-3906	UFLC module remains active after power cycle and Module Nominal Condition has returned. Module nominal condition returned happens when the meter is powered down, therefore no event is recorded, however, if the outage duration is short about 3 seconds, UFLC module remains active on power up. An outage of 6 seconds correctly released the UFLC allowing the next highest RSM module to take control e.g. Ripple.
Mantis-3096	<u>Demand Limit Load Control</u> If Under Frequency Load Control (UFLC) module is triggered due to bad frequency, then the frequency recovers and UFLC starts its Nominal Delay period AND then a short power cycle occurs (about 3 seconds) AND on Power Up the Frequency is still within Normal limits; under these conditions the UFLC module might not correctly exit its Nominal waiting period. This can cause the UFLC module to block other lower priority RSM commands such as Ripple telegrams. It appears that when this incorrect state is reached that it might require two power cycles to correctly reset the RSM logic.
TFS-2246 **TFS-2245** **TFS-6268**	If the U1300 and U3400 meter with R5.0 and R5.4 firmware is running in SSN mode and a backwards time reset over the interval boundary is actioned, only the start time event is recorded in the log.
TFS-2519	<u>Corruption of Previous Billing data when firmware upgrading to R5.0 or R5.4</u> Meters can be F/W upgraded from any R4.x to R5.0 or R5.4. The TOU data and LP data is maintained when the firmware upgrade process expands the TOU area from 8 to 35 days of self-reads. Previous Billing data can be corrupted when F/W upgraded from any R4.x to R5.0 or R5.4. In most applications the Previous Billing data is not used by HES systems or by customers. Re-programming the meter is required to enable the new functionality provided in R5.0 or R5.4 and will also clear the Previous Billing data.

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TFS-1565 **Mantis-3918**	It has been observed that U1300 meters, the LC1 & LC2 random delay closed values tend to have the same minutes and seconds with only the hours being different. Random delay is not random enough for the two relays.
TFS-1067	It is possible that after installation the meter might not switch the LC relay when the first Ripple Telegram is received. If the meter has been programmed with Ripple, then a power outage occurs before the first valid telegram is sent, then the LC relay might not switch. Subsequent ripple telegrams are handled correctly. After the first telegram the meter works correctly. Utilities usually send multiple telegrams even of the same type throughout the day, so this should not have an impact on the consumer.
TFS-1567 **Mantis-3904**	If the Ripple Power Restore option has been set to Time Table AND the random power restore range has been set to 0 then the Time Table schedule will not run. This is unlikely to occur because the random power restore is not likely to be set to 0 by customers.
TFS-11151	U1300 Receiving Ripple Telegrams with full ripple log and a power outage occurs can cause a self-check error and incorrect LP zero fill intervals. This is rare but possible.
Mantis-3254	Season Change event 24 on a U1300 can be logged multiple times in the event log during a partial programming operation.